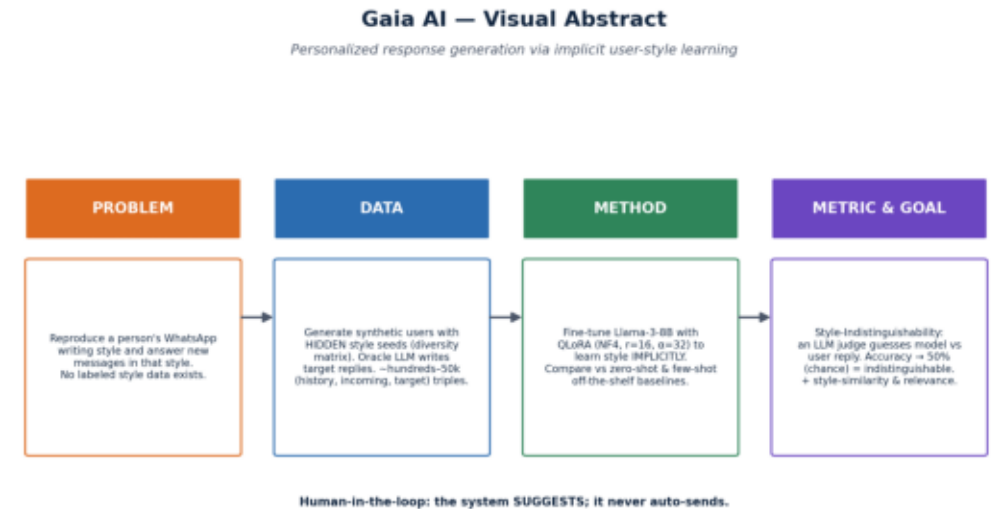


# Gaia AI — Project Proposal

Personalized response generation via implicit user-style learning · Adir · LLM course (HIT)

# 1 · Motivating Use Case

- Communication-heavy people (educators, consultants, founders, support) get floods of messages whose replies follow a consistent PERSONAL style.
- Today this is solved by typing every reply manually, or by generic auto-replies that sound nothing like the user.
  - Why important: time cost is huge and tone consistency matters for relationships/brand.
  - Why hard: a person's style is implicit — sentence length, emoji, slang, punctuation, rhythm — and there is NO labeled 'style' dataset.
- Gaia AI learns the user's style from their own chat history and SUGGESTS replies (never auto-sends — human approves).



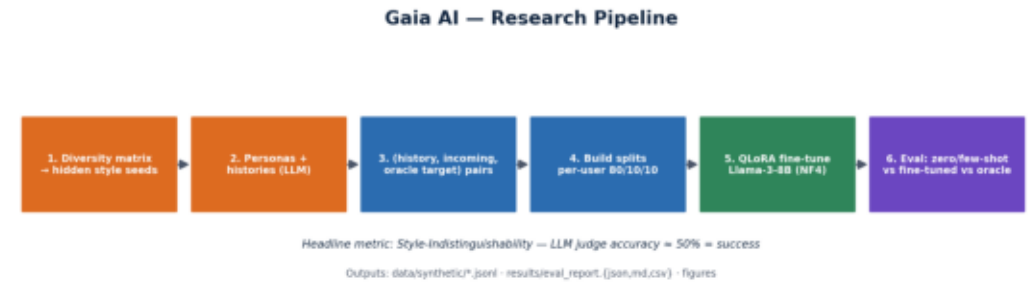
## 2 · Project Task Description

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- Formal problem: conditional text generation with implicit style conditioning.
  - Input: a user's chat history + a new incoming message.
  - Output: a reply that (a) addresses the message and (b) preserves the user's implicit writing style.
- Novelty: learn style IMPLICITLY (no tone labels) and evaluate by Style-Indistinguishability instead of a soft similarity score.
- Scope is novel & data-free: no public dataset exists → we generate synthetic training and test data.

# 3 · Models & Methods

- Pipeline: diversity matrix → synthetic personas+histories → (history, incoming, oracle-target) pairs → splits → fine-tune → eval.
- Fine-tuned model: Llama-3-8B-Instruct + QLoRA (4-bit NF4,  $r=16$ ,  $\alpha=32$ , target q/k/v/o).
- Off-the-shelf baselines to compare against:
  - Zero-shot prompting (style instruction only).
  - Few-shot prompting (5-8 history examples in context).
  - Oracle (LLM that saw the hidden style seed) = upper bound.
- Style memory retrieval (embeddings/Qdrant) provides relevant history at runtime.



## 4 · Data Specification & Generation

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- No real labeled data → generate SYNTHETIC data with an LLM (Self-Instruct-style).
- Each synthetic user gets a HIDDEN style seed sampled from a diversity matrix:
  - language (he/en) × length × emoji × slang × formality × rhythm × punctuation × greeting × occupation × age.
- Per user: ~30–50 style-reference messages + (history, incoming, target) triples; target written by an 'oracle' that sees the seed.
- Splits are PER-USER (no user in both train and test) to test generalization to unseen people.
- Real data (optional, evaluation only): opt-in, anonymized, AES-encrypted, gitignored.

## 5 · Metrics & KPIs

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- **Headline — Style-Indistinguishability:** an LLM judge sees history+incoming+two replies (oracle vs ours, randomized) and guesses the user's.
  - KPI: judge accuracy  $\rightarrow$  50% (chance) = indistinguishable = success.
  - Report 95% Wilson CI + order-bias check; judge model separated from generator.
- **Style similarity:** cosine in a MULTILINGUAL embedding space vs the user's history (he+en).
- **Relevance:** LLM 1-5 — does the reply actually address the message? (style without content is useless).
- **Quality protocol:** per-user split, bias mitigations, ground-truth = oracle (synthetic) / real user (real test).

*Targets: fine\_tuned indist.  $\in$  [45%,55%]; baselines higher (judge can spot them).*